Aegis: structural security at sequence speed

- Sequence-level checks of current screening software easily evaded
- Expensive structure prediction tools
- Screenings that infer **structural context from sequence** required - Aegis bridges that gap
- Looks for toxin-specific interface fingerprints given a sequence
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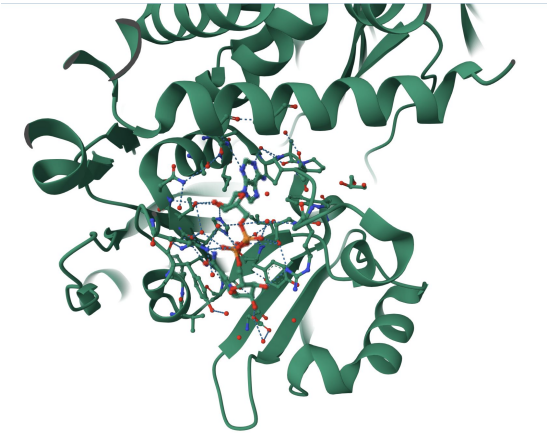
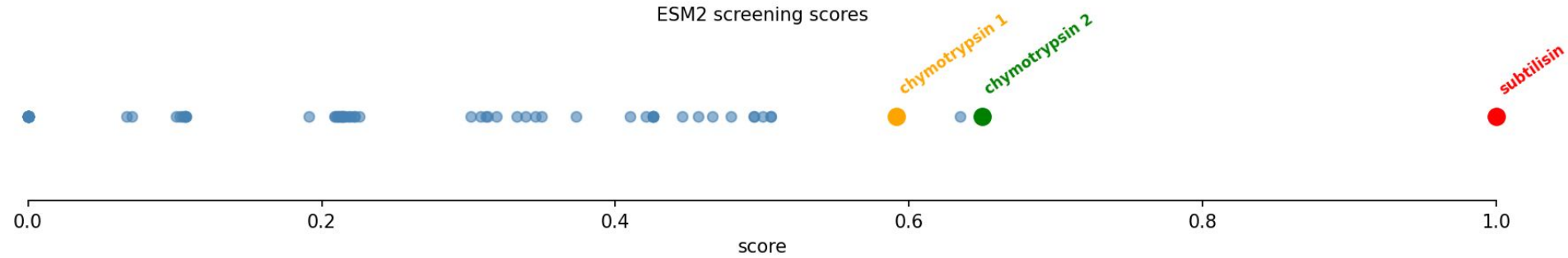


Cholix protein interacting with its substrate NAD

Within your sequence lives an ecosystem.

- Multidimensional vector embeddings assigned by ESM2
- 1. Multidimensional vector embeddings encode the biochemical context of residues (e.g. catalytic serine in an active site)
- 2. Contact maps encode pairwise distance probabilities
- Residue combinations with the right biochemical context (embeddings) in spatial proximity (contact maps) suggest a unique fingerprint for that interface
- Looks for interface fingerprint matches in the sequence